

Distributed LLaMA Inference System

M.Sc Data Science & Artificial Intelligence

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1. Introduction

Large Language Models (LLMs) such as LLaMA require high computational resources and memory capacity for efficient inference. Running large-scale models on a single computer often leads to increased latency, high RAM usage, and expensive GPU dependency. This project proposes a distributed inference framework where multiple connected systems collaboratively execute LLM inference using CPU parallelism over a Local Area Network (LAN). The system follows a Master–Worker architecture where a Root/Master Node coordinates multiple Worker Nodes to distribute computation tasks. This approach reduces memory load per machine, improves inference speed, and enables scalable local AI deployment without requiring costly GPU infrastructure.

2. Problem Statement

Modern LLMs require:

- Large RAM capacity
- High computational power
- GPU-based infrastructure
- Expensive deployment systems

Running larger models on a single machine becomes difficult because of:

- Hardware limitations
- Slow inference speed
- High deployment cost
- Poor scalability

Proposed Solution:

The Distributed LLaMA Inference System distributes model inference workloads across multiple connected machines using CPU parallelism. This creates a low-cost, scalable, and offline AI infrastructure suitable for research and local deployment.

3. System Architecture

The system uses a distributed Master–Worker architecture.

Master Node

The Root/Master Node is responsible for:

- Accepting user prompts
- Coordinating worker nodes
- Synchronizing distributed computation
- Aggregating final outputs

Worker Nodes

Worker Nodes are responsible for:

- Loading model partitions
- Processing inference tasks
- Performing CPU computation
- Returning intermediate outputs

Master Node	→	Worker Nodes
Prompt Handling Task Distribution Result Aggregation		Model Layer Processing CPU Parallelism Distributed Inference

4. Tools & Technologies

Category	Technologies Used
Programming	Python, Bash
Networking	Socket Programming, TCP/IP Communication
AI/ML	LLaMA Models, Local LLM Inference
System Design	Multi-threading, Distributed Computing
Future Deployment	Docker, Kubernetes

5. Models Used

The project was tested using the **LLaMA 2 7B** model.

Features

- Offline inference
- Distributed execution
- Multi-node scalability
- CPU-based deployment
- Low-cost infrastructure

6. Results

The distributed inference system demonstrated significant performance improvements as the number of worker nodes increased.

Benchmark Results

Devices	Total Time	Inference Time
1 Node	1312 ms	1307 ms
2 Nodes	793 ms	739 ms
4 Nodes	494 ms	458 ms

Observed Insights:

- Inference latency reduced significantly with additional worker nodes.
- CPU parallelism improved computational efficiency.
- Lower RAM usage per machine was achieved.
- Better resource utilization was observed across distributed systems.

7. Conclusion

The Distributed LLaMA Inference System successfully demonstrates how large language model inference can be distributed across multiple systems using CPU parallelism. The project provides a scalable and cost-effective solution for local AI deployment without relying on expensive GPU infrastructure. The architecture enables faster inference, lower memory consumption, and improved scalability, making it suitable for research labs, educational institutions, offline AI systems, and distributed enterprise deployments.

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